Comparative Analysis of Ensemble ResNet50 Modalities in Hematological and Dermatological Oncology

1. Abstract

The deployment of deep learning architectures in oncology necessitates high-fidelity diagnostic outputs. This paper evaluates a custom ensemble pipeline utilizing a modified ResNet50 backbone for the dual classification of dermatological lesions (melanoma) and hematological malignancies (leukemia). By leveraging a Medallion Architecture for the underlying data engineering pipeline, the model achieves production-grade inference latency while maintaining clinical accuracy.

2. Algorithmic Methodology

To combat the inherent class imbalance present in early-stage oncological imaging datasets, standard cross-entropy is insufficient. Instead, the ensemble relies on a dynamically weighted Focal Loss function, allowing the model to focus on hard-to-classify, sparse examples (e.g., differentiating atypical nevi from early-stage melanoma).

The loss function is mathematically defined as:

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

Where p_t represents the model's estimated probability for the ground truth class, α_t is the weighting factor for class imbalance, and γ is the focusing parameter that smoothly adjusts the rate at which easy examples are down-weighted.

3. Dataset and Bronze-to-Gold Pipeline

Raw DICOM and high-resolution dermoscopy images are ingested into a Bronze layer object store. Transformation logic extracts metadata and normalizes pixel intensities, promoting the data to the Silver layer. The Gold layer contains tensor-ready, augmented arrays used directly for distributed model training.

[Insert Dummy Image Here]

Figure 1: Visual representation of the Bronze-Silver-Gold transformation pipeline alongside the ResNet50 feature extraction layers.

4. Clinical Evaluation Metrics

The ensemble model was evaluated against isolated baseline models. The results indicate a statistically significant reduction in false negatives, a critical metric in oncology.

Table 2: Efficacy Metrics Across Oncological Domains

Target Pathology	Architecture	Sensitivity	Specificity	AUC-ROC	False Negative Rate (FNR)
Melanoma	Baseline CNN	0.82	0.88	0.89	0.18
Melanoma	Ensemble ResNet50	0.94	0.91	0.96	0.06*
Leukemia (ALL)	Baseline CNN	0.79	0.85	0.84	0.21
Leukemia (ALL)	Ensemble ResNet50	0.91	0.93	0.95	0.09

**Note: The FNR reduction in the dermatological pipeline was achieved by aggressive hyperparameter tuning on the γ variable within the focal loss function, explicitly penalizing missed malignant classifications.*

5. Conclusion

The integration of specialized loss functions within robust, containerized MLOps pipelines demonstrates that clinical-grade diagnostic tools can be successfully transitioned from isolated research environments to scalable, backend inference engines.